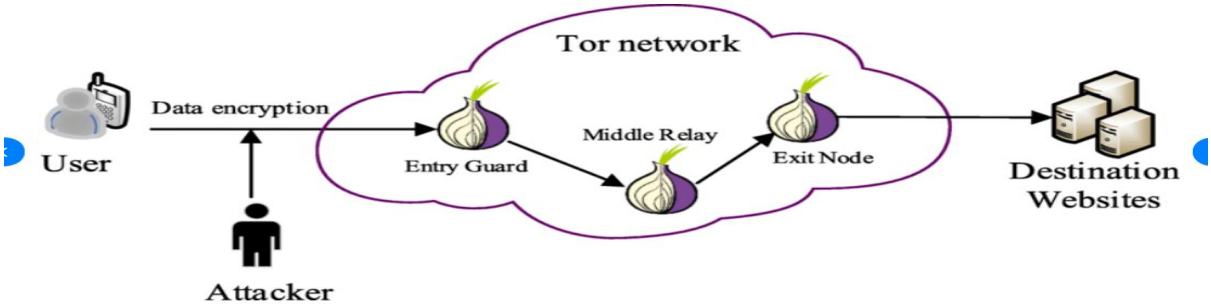
ANONYMITY

Staying Anonymous: Anonymity means keeping your identity unknown on the Internet. When fully implemented, it is totally difficulty to identify an entity.

To remain anonymous, you must prevent your digital activity from being linked to your real-world identity by using tools that mask your IP address, encrypt traffic, and prevent tracking. Key methods include using a trusted VPN, the Tor browser, secure messaging apps like Signal, and operating systems designed for privacy, such as Tails.

Tools : Tor, Proxychains, MAC-Changer, Nipe, Duck Duck Go, etc

TOR NETWORK: The Tor network (The Onion Router) is a free, open-source system designed for online anonymity and privacy, routing internet traffic through a global, volunteer-operated network of over 7,000 relays. By encrypting data in multiple layers and passing it through three random nodes, it masks a user's location and usage, making it difficult to trace or surveil.



A Tor anonymity network example showing a user connecting to the Internet via three Tor nodes. The WF passive attack occurs between the user and the Tor entry guard

Key Aspects of the Tor Network:

- **Onion Routing:** Data is wrapped in multiple layers of encryption (like an onion), with each relay in the circuit decrypting one layer to find the next destination, ensuring no single node knows the entire path.
- **Purpose:** Originally developed by the U.S. Naval Research Laboratory, it now serves journalists, activists, and individuals seeking to bypass censorship or surveillance

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- **Accessing Tor:** Users primarily access the network using the Tor Browser, which is pre-configured to connect safely.
- Onion Services **(.onion):** Websites can operate within the Tor network, providing hidden services that protect the identity of both the site operator and the visitor.
- **Limitations:** While it offers high privacy, browsing through the Tor network is generally slower than conventional internet usage due to the multiple relay hops.

Benefits and Uses:

- **Anonymity:** Hides your IP address and browsing activity.

- **Censorship Circumvention:** Bypasses blocked sites and network restrictions.
- **Privacy:** Protects against, tracking, and surveillance.

Tor Project +2

The Tor Project maintains the network, relying on a community of volunteers to run the node

Proxychains is a UNIX tool (often used in Kali Linux) that forces any TCP application's connection through a series of proxies (SOCKS4/5 or HTTP), masking your original IP address for enhanced anonymity, bypassing firewall restrictions, or conducting secure, indirect network scans. It supports dynamic, strict, or random chain chaining options and is commonly used with Tor.

Key Features and Details

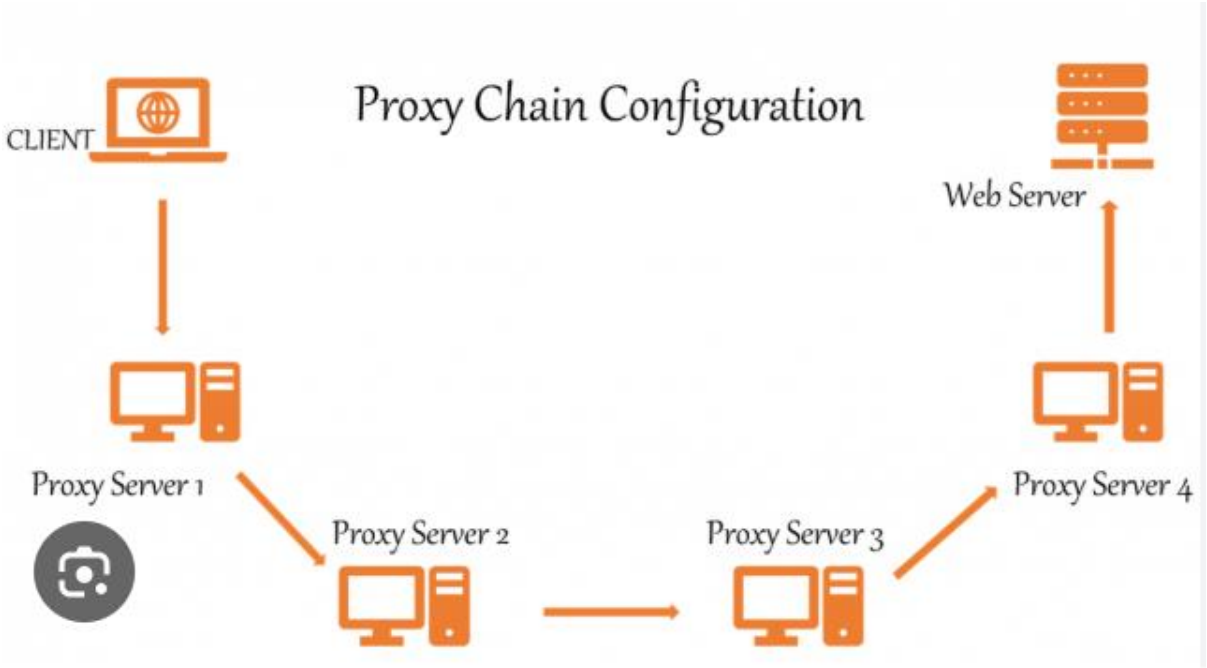
- **Functionality:** It hooks network-related libc functions in dynamically linked programs to redirect traffic.
- **Supported Proxies:** SOCKS4, SOCKS5, and HTTP/HTTPS
- **Usage:** Configured via /etc/proxychains.conf or /etc/proxychains4.conf, where you list proxies in the desired order.
- **Commands:** Run applications through it using proxychains <command> (e.g.proxychains firefox).
- **Limitations:** Supports TCP only; does not support UDP or ICMP (e.g.ping will not work).

Common Use Cases

- **Anonymity:** Hiding your IP address from destination servers.
- **Bypassing Restrictions:** Evading firewalls or geo-restrictions.
- **Security Testing:** Anonymous scanning and penetration testing

Types of Chaining Modes

- **Dynamic Chain:** Proxies are used in the order listed, but dead proxies are skipped.
- **Strict Chain:** Proxies are used in the exact order listed; if one fails, the entire connection fails.
- **Random Chain:** Proxies are chosen randomly from the list.



NIPE TOOL: **Nipe** is a program that uses the Tor network as the user's default gateway, routing all traffic on the Tor network, which is often used to provide privacy and anonymity. It should be emphasized that hiding an IP address alone will not provide anonymity when using a tool for privacy and anonymity, as DNS information may still be exposed. Both IP and DNS information must be hidden.

STEPS

1. Check the Internet Status
   Ping jntuh.ac.in -c 3
2. Install TOR
   Apt -get install tor -y apt install tor -y
3. Configuring Proxychains
   - Open proxychains4.conf file
   - Nano /etc/proxychains4.conf
   - Uncomment “dynamic Chain”
   - Comment “Strict Chain”
   - Uncomment “DNS Leak”
   - Add socks5 and enter the following IP & Port : 127.0.0.1 & 9050
   - Save the file
4. Ifconfig, note the IP
5. Macchanger macchanger -s eth0
6. Change MAC address macchanger -r eth0
7. Start the TOR service
   Service tor start

8. Check the TOR status
   Service tor status (Check for active in green color)
9. Open the browser check whatismyip.com
10. Implement proxychains
    Proxychains4 firefox www.duckduckgo.com (anonymous engine)
11. Open the browser check whatismyip.com
12. xxx